When is a cheap rung safe to trust? Rung bias and a pilot-checkable bound for multi-fidelity search over open-source chip flows

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***Abstract*—Open RTL-to-GDS flows put a place-and-route run within reach of a laptop, but a search over ABC pass sequences needs hundreds of runs, so practitioners rank candidates on a cheap rung (generic-gate count and depth, liberty-mapped synthesis, placement) and trust it. We measure how far that trust is justified on SKY130: 10 small designs, 60 ABC pass sequences each, two clock targets, four rungs, 1,200 runs, with routed area x delay as reference. The generic-gate rung is uninformative to anti-correlated with the routed ranking (median Kendall tau-b -0.04 at the tight clock); mapped and placed rungs mis-rank at a tight clock (median pick regret 9 to 10.5%) and are almost exact at a relaxed one. Replayed at equal wall-clock, cheap-rung screening and plain successive halving do not beat random or hill-climbing search on this table (about 10% against 0.9% mean routed regret at 45 s). We give a three-line worst-case bound (regret of halving is at most the sum of the cut rungs' *inversions*), show it is vacuous for the generic rung, and test pilot-checkable substitutes empirically: a statistic built from the pilot's best three points against the rest covers the realised regret in all 40 group-start cells at a 16-point pilot (minimum 0.81) but is informative only where the rung is already faithful. A conformal shortlist on the mapped rung retains the optimum 0.99 of the time at nominal 0.90 while routing 42 to 57% of the pool; on the generic rung it is no better than random. A proof-success predictor is not supported.**

Draft for MLCAD (6 pages when typeset); not submitted. Programme `verified-dse`, milestone 7. Written with AI assistance (Claude Code). Every number below is copied from a committed results file named beside it; `sh reproduce.sh` regenerates them from the cached flow tables. No affiliation with any lab is claimed. Submission is an owner step.

A. 1 Introduction and related work

Multi-fidelity search is mature for hyperparameters: GP bandits with known maximum bias (Kandasamy et al., 2016, arXiv:1603.06288), max-value entropy search over fidelities (Takeno et al., 2019, arXiv:1901.08275), successive halving (Jamieson & Talwalkar, 2015, arXiv:1502.07943), Hyperband and BOHB (arXiv:1603.06560, arXiv:1807.01774), with BoTorch as software (arXiv:1910.06403). Chip-flow search papers mostly use one fidelity: ABC pass-sequence learning (Yu, Xiao & De Micheli, 2018, arXiv:1804.05714), surveyed in Huang et al. (2021, arXiv:2102.03357). Mirhoseini et al. (2021, doi:10.1038/s41586-021-03544-w) and its reassessment (arXiv:2302.11014) are the reason we use equal-budget baselines. Conformal prediction (Angelopoulos & Bates, 2021, arXiv:2107.07511) supplies the distribution-free tool for shortlists. The related-work claims rest on titles and abstracts returned by `research/_shared/cite_check.py`, not on a full reading of every paper (`PROPOSAL.md`). The inversion bound of Section 4 may be a special case of known successive-halving analyses; we do not claim it as new.

B. 2 Setup

Designs: 10 small combinational designs plus one FIFO (`designs/`). Space: 60 ABC sequences of length 0 to 3 over `balance`, `rewrite`, `rewrite -z`, `refactor`, `refactor -z`, `dc2`, `dch`; 10 fail in ABC and are dropped, leaving pools of 50 or 52. Rungs: R1 generic gates (cells x depth, 0.15 s per point), R2 liberty-mapped without wires (0.8 s), R3 after placement (1.0 s), R4 routed with `repair_timing -setup` (1.3 s), all one core. Score: area x period; clock targets are each design's own no-wire baseline (tight) and 1.5x that (relaxed). Flow: Yosys 0.33, OpenROAD 26Q2, SKY130 HD, one corner (tt 25C 1.8 V), 40% utilisation. "Routed" is global-route-estimated, not detail-routed.

C. 3 Results

Rung bias (`experiments/rung_bias/results.txt`). R1 tau-b against routed: -0.04 tight, -0.01 relaxed; the 95% bootstrap interval lies below 0 for popcount16 and crc16 at the tight clock. R2: 0.59 / 0.73; R3: 0.50 / 0.76. Median pick regret tight / relaxed: R1 13.5% / 20.0%, R2 10.5% / 0.0%, R3 9.4% / 0.0%. Timing repair, seen only by the later rungs, reorders designs at the tight clock.

Search at equal budget (`experiments/search_baselines/results.txt`, `experiments/halving/results.txt`; 400 paired seeds, replayed on the cached table). Mean routed regret at the tight clock: random R4 6.3 / 2.1 / 0.94% and hill climbing 7.4 / 1.7 / 0.61% at 10 / 30 / 45 s; screening all points at R1 then verifying the top 15.0 / 5.3 / 1.08%. Plain halving from R1 is the worst strategy at every budget of 10 s or more. A 6-point routed pilot choosing the starting rung recovers the designs plain halving loses (alu8 16.9% to 0.8%), but most of that gain is "skip R1". Net: no multi-fidelity gain at equal wall-clock yet; routed runs here cost only about 8x the cheapest rung, so there is little to save, and the pool caps halving beyond roughly 30 s.

Conformal shortlist (`experiments/conformal/results.txt`; 20 design-clock groups). Fit routed value from a rung on a 10 to 16 point routed pilot, keep points whose one-sided conformal bound could still beat the pilot's best. R2: optimum

retained 0.99 at nominal 0.90 (0.90 to 0.92 at 0.80) while routing 42 to 57% of the pool, against 0.53 to 0.65 for random search with the same routed count. R1: 0.65 at nominal 0.80, no better than random, so the hypothesis fails for R1.

Proof success (`experiments/proof_success/results.txt`; 752 SAT-miter proofs, 15 designs, 10 s timeout). Rewrites never failed a proof; time is set by the design (median 0.01 s at 4-bit multiplier, 11 s at 8, none of 100 proved at 9 and 10 bits). Pooled leave-one-design-out AUC 0.96 to 0.998 equals that of an oracle knowing only each design's success rate. Within the one varying design (8-bit multiplier, 17 of 50 proved) AUC is 0.65 to 0.78; "the rewrite is smaller than the reference" gets 0.81. We do not claim proofs are predictable from cheap features.

Proof success under Yosys 0.70 (milestone 52, `experiments/proof_success/compare_y070.txt`; 900 launched, 751 usable, 751 shared with the Yosys 0.33 table). 0 counter-examples; 627 proved and 124 timed out at 10 s. Every design except the 7- to 10-bit multipliers proves 50 of 50 within 1 s on both versions, the 10-bit multiplier times out on all 50 under both, and the proved-within-3-s label agrees on 750 of 751 points (0.999). So the reading that proof time is set by the design is not an artefact of the old solver. One more sequence failed in ABC under 0.70 (751 usable points against 752), not investigated, and mapped netlists differ by a few cells between versions.

Other designs, corners and operating points (milestones 9, 15, 24 to 28; each paragraph names its results file). *Larger designs* (`experiments/large/`, `experiments/nonarith/`): in ten designs of two families (multipliers and adders; shifters, popcounts and an ALU) the R2 cut floors on one or two designs per family at the relaxed clock (mult24 and mult32 at 11.3% and 14.1%; barrel64 in the second family), and skipping the R2 cut removes the floor. *Placement density* (`experiments/util65/`): at 65% utilisation instead of 40% the median R2 regret over the five larger designs moves by 0.4 points relaxed (2.9% to 2.5%) and 1.2 tight (0.8% to 2.0%); a routed point costs 24.4 s instead of 19.4 s. *Second corner* (`experiments/corner_ss/`, `experiments/corner_replay/`; liberty only): at SKY130 ss 100C 1.60 V the routed-ADP ranking has a median Spearman 0.69 against tt, the tt-optimal sequence loses a median 3.4% (up to 27%), and chosen-eta halving beats random R4 at the relaxed clock but not at the tight one from 30 s, the reverse of tt, so a halving schedule has to be checked at the corner being signed off. *Held-out families* (`experiments/unseen/`): CRC-32, priority encoder, sorting network, S-box and decoder-mux have no tight-clock cell with R2 regret above 10% (largest 9.3%), and fixed eta = 1.5 beats the no-cut policy f3 there (0.77 against 0.85% at 25 routed-point equivalents), so f3 is insurance, not a universal winner. *Designs written to break R2* (`experiments/hostile/`; crossbar, GF(2) matrix-vector, broadcast mix, 128-bit comparator, rotate-mix): one design (`bcastmix192`, two distinct outcomes over 60 sequences) has R2 regret 49.9% because repair buffers the mapping that looked worst (tau(R2,R4) = -1); the other eight cells are at most 1.1%. f3 (0.67) again matches the best arm at 25 equivalents and the best pilot gate is worse (1.05); the pilot's AUC of 1.00 rests on two cells of one design. Across both new sets, the pilot gate never paid for its pilot. Pooled over the ten held-out designs (`experiments/heldout_signif/`), eta = 1.5 and f3 beat random R4 by 1.3 points at 25 equivalents (8 and 7 of 10 designs better, bootstrap intervals exclude zero), but no test survives a Holm correction over 12 tests (smallest adjusted p 0.14), and f3 and eta = 1.5 do not differ at 25 (p 1.0); the held-out win is reported uncorrected. Adding the two high-fanout designs (`fanao96`, `bcastadd64`; twelve designs) and ranking the R2 rung by mapped area alone instead of area x period (`experiments/r2_area_arm/`) changes nothing inside halving: the two scorings differ by 0.00 points at 25 equivalents (Holm p 1.0), because eta = 1.5 keeps two thirds of the points after R2. As an argmin, area alone is worse (mean full-pool R2 pick regret 10.3% against 6.0% over 24 cells) although its mean tau with R4 is higher (0.58 against 0.38). On twelve designs both halving arms beat random R4 by 1.2 points at 25 equivalents (10 of 12 designs better, Holm p 0.044), the first held-out comparison to survive a 12-test correction; two of the twelve were added after the ten-design test. A harsher first cut (eta = 3, `experiments/r2_harsh_cut/`) shows the R2 score matters there: mean regret at 25 equivalents is 4.67 points against 0.65 at eta = 1.5 and 1.86 for random R4, and does not fall with budget, but the mean is two designs (`bcastmix192`, 49.9, a two-outcome binary swap; `crc32`, 2.9). The median over twelve designs is lower at eta = 3 (0.31, against 0.76 and 1.21), so a harsh cut is the best typical case and the worst tail; none of this is significant after correction (Holm p 1.0). Scoring R2 by area alone is worse still at eta = 3 (5.94 at 25). The medians and the exclusion are post hoc.

Tail table (`experiments/tail_risk/`, descriptive, fixed before the run): worst-design regret of eta = 1.5 halving falls with budget (8.3 / 4.9 / 1.4 / 0.6% at 10 / 15 / 25 / 40 equivalents) and no design exceeds 5% from 15 equivalents up; at eta = 3 the worst design is `bcastmix192` at 47 to 50% at every budget (mapped-area scoring: `bcastadd64`, 61 to 68%), while its median is lower from 15 equivalents (0.85 against 1.87 at 15). Random R4 has four designs above 5% at 10 and 15 equivalents. We therefore report medians and worst-design regret beside means.

Detail routing of the larger designs (milestones 78 to 80, `experiments/detail_route/`, moved here from the to-do list). Detail routing on mult24 and mult32 (milestone 78 measured one mult24 point at 877 s of detail routing on one core, 26 times mult16, so a full cell of 50 points is about 12 core-hours and was not run; milestone 79 detail-routed 12 mult24 relaxed points spread over R4 rank: tau-b(R4,R5) 0.73, R5-best point is R4's best, R2 pick regret 0.3%, median 1210 s per point, 0 DRC; milestone 80 did 4 mult32 relaxed points: R4 best is R5 best, R4 pick regret 0.0%, R2 1.8%, median

1213 s per point, 0 DRC; milestones 47, 49 and 51 did five small designs at both clocks and mult16 and the 64-bit adders at the tight clock), and the ss replay of the larger designs (liberty-only corner so far; needs OpenROAD).

Fanout mechanism (milestones 30 and 31, `experiments/fanout_rich/`; the causal sweep is milestones 73 to 77 and Section 5). The mechanism on `fanao96` is now observational (milestone 31: area keeps its order, tau 0.98; the mapped period spread of 0.58 to 3.88 ns compresses to about 1.1 to 2.1 ns routed, and buffers added do not explain it); a causal test varying the fanout is still open (milestone 30, `experiments/fanout_rich/`: 36 to 39 distinct outcomes per cell, R2 pick regret 13.7 and 14.6%, tau 0.14 and 0.20, while the gate still does not beat the no-R2-cut arms); `bcastmix192` alone has two outcomes and supports no statistics.

Model-based baselines (milestones 20 to 23, 54 to 67, `experiments/bohb/`, `experiments/mf_surrogate/`; moved here from the to-do list, text unchanged). Bayesian-optimisation baselines at equal wall-clock on larger pools (ridge, GP-EI and TPE variants on pools of about 50 are done, milestones 20 to 23, and none beats never cutting on R2; BOHB with KDE entrants inside Hyperband brackets, milestone 54, `experiments/bohb/`, is the exception at low budget: mean regret 2.62 and 2.05 % at 15 and 25 routed-run equivalents against 3.56 and 1.77 % for random routing, ahead in 26 and 25 of 40 cells (sign-test p 0.011 and 0.029, non-independent cells), and behind it at 40 (1.38 against 0.36 %), with the same tail cells (barrel64, mult32) at 14 to 16 %; exploratory; milestone 55, `experiments/bohb/results_unseen.txt`, replays it on the five held-out designs of milestone 27 with the predictions fixed beforehand: mean regret 0.64, 0.49 and 0.14 % at 15, 25 and 40 equivalents against 2.75, 1.30 and 0.32 % for random routing, ahead in 8 of 10 cells at each budget (5 designs, not independent, so counts and not evidence). Two of three predictions held; the third was wrong, since BOHB also beat random at 40. These designs are R2-faithful, so the direction replicates and the size may not; its worst held-out cell is crc32 tight (3.4, 2.8, 0.8 %), where R2 is least faithful, and a held-out set with an unfaithful R2 is still untested). Milestone 57, `experiments/bohb/sweep.txt`, varies gamma, random fraction and minimum observations one at a time over all 50 cells: the 15-equivalent win on the pooled mean holds in 8 of 8 settings (2.02 to 3.14 % against 3.40 % for random routing) but a minimum of 10 observations nearly removes it (3.14 %), and at 25 equivalents BOHB is below random on the pooled mean in only 4 of 8 settings (the defaults give 1.74 against 1.68 %), so the 25-equivalent claim rests on per-cell counts, not the mean; random routing stays ahead at 40 in every setting. Milestone 58, `experiments/bohb/minobs_grid.txt`, refines the minimum-observations axis (3, 5, 8, 12): at 15 equivalents BOHB is below random (3.40 %) up to a minimum of 8 (2.16 to 2.59 %) and the win fades gradually, not at a sharp threshold, by 12 (3.28 %), with a shallow optimum near 5 to 6; no setting gives a 25-equivalent pooled-mean win above 0.2 points and random stays ahead at 40. The settings were scored on the cells they were chosen from, so no new default is proposed. Milestone 60, `experiments/bohb/oos_minobs.txt`, scores the choice out of sample: a minimum of 5 chosen on the 40 main cells gives 0.64 % on the 10 held-out cells at 15 equivalents against 0.42 % for the held-out best (3) and 2.75 % for random routing, so the choice transfers; BOHB is also below random on the held-out cells at 25 and 40 (a third prediction, that random would be level there, was wrong), while on the 40 main cells random wins at 40, so the size of the BOHB advantage depends on the design set. Milestone 61, `experiments/bohb/faith_corr.txt`, tests the explanation offered for that difference, that R2 is more faithful on the held-out designs: across the 50 cells the Spearman correlation between the cell's Kendall tau of R2 against routed and BOHB's gain over random routing is +0.19 at 15 equivalents (95% interval -0.13 to +0.48), +0.14 at 25 and +0.08 at 40, so the data neither show nor exclude a modest link and the explanation stays a hypothesis. Milestone 62, `experiments/bohb/degrade.txt`, tests it causally by corrupting R2 with synthetic noise: mean tau falls from 0.62 to 0.06 yet BOHB's 15-equivalent regret only rises from 2.62 to 3.40 % (random routing 3.56 %), so most of its low-budget advantage does not depend on R2 being faithful (iid noise; structured unfaithfulness untested). Milestones 63 and 64 (`degrade_score.txt`, `degrade_features.txt`, `degrade_none.txt`) separate the channels: corrupting only the R2 score or only the KDE features at sigma 1 leaves 0.6 to 0.8 of the 0.94-point 15-equivalent win (2.73 and 2.94 % against 3.56 %), and removing R2 information entirely (constant scores, noise features) gives 3.79 % at 15 equivalents, 0.23 points worse than random routing (milestone 66, `degrade_paired_none.txt`: paired bootstrap over the 40 cells [-0.23, +0.71], level in 18 cells each way, so indistinguishable from random), and 2.19 against 1.77 % at 25 (+0.42 [+0.12, +0.71], sign test p 0.011) and 1.25 against 0.36 % at 40 (+0.89 [+0.60, +1.19], $p < 0.001$, worse in 34 of 36 non-tied cells): with no R2 signal BOHB is no better than random at 15 and reliably worse at 25 and 40. So the edge needs some R2 signal, though very little (tau 0.06 still beats none by about 0.4 points); an earlier guess that the routed observations and bracket structure carry it is not supported. The same paired test on the other arms (milestone 67, `degrade_paired*.txt`, bootstrap over 40 non-independent cells, descriptive) puts the 15-equivalent gain over random at -0.94 [-1.72, -0.19] points with a faithful R2, -0.83 [-1.48, -0.21] with score noise sigma 1, -0.63 [-1.24, -0.04] with feature noise sigma 1, and -0.16 [-0.76, +0.45] with both: only the both-corrupted arm is level with random. At 40 equivalents BOHB is worse than random in every arm including the faithful one (+1.01 [+0.22, +2.06]). All on iid synthetic corruption; structured unfaithfulness needs the flow.

BOHB when R2 is structurally unfaithful,

and gates that did not help (milestones 68 to 71, `experiments/unfaithful/`, `experiments/bohb/results_unfaithful*.txt`, `gate_*.txt`, `gate3_*.txt`; 100 paired seeds; OpenROAD 26Q3 and Yosys 0.33 for the new flow runs, earlier tables 26Q2, a confound). Four new multiplier widths (20, 28, 36, 40; 60 sequences from seed 0, both clocks, 80 of 480 runs failed in ABC) were run under the milestone 9 protocol. At the relaxed clock three of four cells (mult28, 36, 40; tau(R2,R4) 0.57, 0.41, 0.65) leave every rung-cut method stuck: BOHB keeps 6.9, 6.5 and 8.5% regret at 25 and 40 equivalents, equal to successive halving from R2, while random routing reaches 0.2, 0.1 and 0.9% and BOHB routing first (never cutting on R2 alone) recovers. Tight-clock cells are at or below 1%. So BOHB's low-budget edge (milestones 54, 55, 57) comes with a tail in a clock regime a practitioner would use; the failing regret grows with width; this is one family chosen as the bad one, four cells, no pooled test. Two gates were tried over 48 cells (the 40 main cells plus the 8 multiplier cells), fixed in advance and untuned: six routed probes with a Kendall-tau threshold of 0.4 that falls back to routing-first (mean regret 3.14, 1.42, 1.16% at 15, 25, 40 equivalents against BOHB 2.39, 2.17, 1.62 and random 3.41, 1.67, 0.35; it halves the multiplier tail, 5.88 to 3.27% at 40, but falls back on only about a third of failing draws and a fifth of healthy ones), and six placed probes comparing R2 with R3 at no routing cost (3.81, 2.60, 1.98%; worse than BOHB in 34 of 40 main cells, since placement inherits R2's ordering and the failure arises in timing repair). Neither beats the simple rule that fits the table: with 25 or more routed-run equivalents, do not cut on R2 first; BOHB is a 15-equivalent tool.

D. 4 A bound and what a pilot can check

Define the inversion of a rung, iota = max(0, max over pairs with s(x) <= s(x') of log y(x) - log y(x')). **Proposition.** Successive halving with cut rungs r_1..r_m returns a value with log regret at most the sum of iota_{r_j}. Proof: if the best survivor x of cut j-1 is cut, every survivor z has s(z) <= s(x), so log y(z) <= log y(x) + iota; chain over the cuts (`experiments/theory/BOUND.md`; checked on 500 random instances and all 40 cells). It is vacuous for R1 (iota above 0.3 on 8 of 10 groups; bound 0.63 to 0.71) and loose for R2 (bound 0.23 to 0.35 where realised regret is about 0). A pilot's maximum is a lower bound on iota, so a pilot can suggest but not certify it.

A bound that uses the survivor count (milestone 37, `experiments/theory/KTH.md`, `kth_results.txt`). Let iotaˆ(k) be the largest, over candidates x, of the k-th largest excess log y(z) - log y(x) among rivals z the rung ranks at or ahead of x. A cut keeping k points loses at most iotaˆ(k) (x is cut only if all k survivors are such rivals, and the best of any k of them is at most the k-th largest), so regret is at most the sum of iotaˆ(k_j); iotaˆ(1) = iota. On the 40 cached group-rung cells the bound is never violated (200 tie-breaks each), is below the old bound in 39 of 39 cells with a positive old bound (median ratio 0.54), and is under the spread of y in 32 cells against 16. It remains about 3 times the largest realised regret and needs the full table. Anchoring the first cut at the pool optimum (milestone 39, `anchored_results.txt`) is also valid (0 violations) but lowers the bound by only about 11% at the median (lower in 23 of 37 cells).

A rank assumption for zero regret (milestones 44 and 45, `experiments/theory/optimum_rank_results.txt`, `optimum_rank_n_results.txt`; no flow run). Let rho_j be the number of rivals a cut's score ranks at or ahead of the pool optimum x. *A cut keeping k_j points can only drop x* if rho_j >= k_j, so rho_j < k_j at every cut makes regret exactly 0 (the kappa = 0 case of the bound above; no violation in 200 tie-breaks of each of the 11 guaranteed cells). The assumption holds in 11 of 40 cached cells (cells whose worst rho/k is at most 1, 2 and 4: 13, 21 and 29 of 40; median 1.97), and where it fails 0.167 of tie-breaks still reach zero regret on average. Subsampling the pool to n' = 20, 30, 40 and 50 candidates, the median worst rho/k rises 2.33, 2.60, 3.00, 3.00 and the share with rho/k at most 1 falls 0.168, 0.130, 0.086, 0.050; this is partly built in (the last cut keeps one point, so its ratio is the raw count rho_3), and nothing above 50 candidates was measured. It is a diagnostic on the full table, not a pilot-side certificate: a pilot cannot observe rho, and milestone 43 shows a quantile of the excess cannot supply the missing bound.

Pilot-checkable statistics (`topk_results.txt`, `sweep_results.txt`; 200 to 300 pilots x 10 to 20 tie-breaks per cell, 20 groups, both halving starts). Max over random pilot pairs undercovers (under 0.8 in 6 of 40 cells, worst 0.28). The top-3-vs-all statistic covers in all 40 cells at m = 16 (minimum 0.81) but is a median 4.6 times the realised regret and under 0.1 only for halving from R2 at the relaxed clock (7 of 10 groups). Across pilot sizes the share of regret-bearing cells covered is 0.855 / 0.915 / 0.969 / 0.988 / 0.995 at m = 6 / 8 / 12 / 16 / 24, with median bound 0.21 to 0.33. Coverage far exceeds the probability that the pilot holds one of the pool's best three (0.32 at m = 6, matching the hypergeometric formula), so that event is not the mechanism. The coverage is empirical, not proved.

A pilot gate (milestone 16, `experiments/pilot_gate/results.txt`; all 20 designs, both clocks, budgets of 10 to 40 routed-point equivalents, 400 paired seeds). A five-point pilot's tau(R2,R4) separates the 12 of 40 cells where cutting on R2 rank misses by over 10% from the rest with AUC 0.85 (0.74 using R2 against R3), but a gate that picks halving with eta = 1.5 or the no-R2-cut arm from that statistic is within 0.1 points of the simple no-R2-cut arm at 25 and 40 equivalents (1.31 and 0.34 against 1.40 and 0.44; random 1.74 and 0.39) and worse at some budgets. On 20 designs eta = 1.5 loses to random at 25 and 40 equivalents (2.12, 1.38) because of barrel64; an oracle gate would reach 0.98 and 0.04. Six gates were compared without correction.

Detail routing and tool versions (milestone 47, `experiments/detail_route/results.txt`; 5

designs, 250 points, tight clock, OpenROAD 26Q3 and Yosys 0.70). Against a detail-routed reference (detailed_route plus OpenRCX extraction), R4 (global route plus repair) has tau-b 0.65 to 0.93 (median 0.84, three of five designs at 0.8 or more), its pick costs at most 1.2% and the R5 optimum is in its top 5 on all five designs; R2's bias at the tight clock is unchanged (tau 0.25 to 0.86 against 0.36 to 0.85 for R4, pick regret 10.1% on alu8). Extracted slack is 0.08 to 0.35 ns more optimistic than the global-route estimate. Re-running R4 under the newer tools gives tau of only 0.28 to 0.56 against the cached table, so numeric regrets in this paper hold for OpenROAD 26Q2 and Yosys 0.33; the rung-bias conclusion survives, the tables do not transfer. One corner, five designs, the version and reference changes are confounded in the drift row. **Relaxed clock** (milestone 49, `experiments/detail_route/results_relaxed.txt`; the same five designs and 250 points at 1.5x the baseline period, 0 DRC violations): R4 against R5 has tau-b 0.81 to 0.90 (median 0.85, all five at 0.8 or more), its pick costs 0.0% on four designs and 2.3% on mult8, and the R5 optimum is in its top 5 on all five. R2 against R5 has median tau 0.66, with pick regret 10.1% on alu8 and 9.2% on barrel16 (0.0% at the tight clock), so R2's bias on barrel16 is clock-dependent. Extracted slack is a median 0.325 ns more optimistic than the global-route estimate (tight: 0.19 ns), and version drift of R4 remains large (tau 0.07 to 0.66). No predictions were fixed before this run. **Larger designs** (milestone 51, `experiments/detail_route/results_large.txt`; mult16 and the 64-bit adders `adder_op64` and `adder_ripple64`, tight clock, first 24 sequences per design, 71 of 72 points ok, 0 DRC violations; the predictions were written before the results were read): R4 against R5 has tau-b 0.93 on mult16 and adder_op64 (the ripple adder, 1.00, has only 6 distinct R5 scores and is uninformative), R4's pick costs 0.0% on all three and the R5 optimum is in its top 5 on all three. R2 against R5 has tau 0.80 (adder_op64) and 0.67 (mult16) with pick regret 12.0% and 3.9%. Extracted slack is more optimistic than the global-route estimate on all 71 points (median 0.39 ns). Version drift of R4 against the cached table is again large on the multiplier (tau 0.31). Pools of 24 give wide intervals; mult24 and mult32 were not run (220 to 280 s per mult16 detail route).

Observable rank displacement (milestone 84, `experiments/theory/pilot_rho.py`, `pilot_rho_results.txt`; no flow run). Rung scores are known for the whole pool, so the rank-displacement ratio of milestone 44 is exact for every routed pilot point. Let T be that ratio for the pilot's best routed point. Over the 40 cells T is at least the unrouted optimum's ratio in a median 0.89 / 0.97 / 0.99 of pilots at m = 8 / 16 / 24 (27, 27 and 28 cells at 0.8 or more); calling T < 1 a certificate marks 0.70 of pilots at m = 16 in the 11 cells where the optimum meets the zero-regret condition, and 0.039 of pilots in the 29 where it does not. Empirical on this table, not a theorem.

Pilot-side certificate and its limits (milestones 35 to 43, moved here from the to-do list). A proof or counterexample for the top-vs-all coverage, or a conformal pilot statistic with a theorem (partial, milestone 35: the pair-quantile share bound is proved only for independently drawn pairs; for disjoint pairs the iid argument fails in general, counterexamples at N <= 7, `experiments/theory/disjoint_check.py`, and none was found at the claim's threshold; it is still a share bound, not a regret bound; milestone 36, `experiments/theory/with_replacement_results.txt`: the independent-pairs pilot, which the proof covers, has empirical coverage 0.826 or more over 60 cells at about 14 distinct routed points; milestone 38, `kth_pilot_results.txt`: a 16-point pilot estimate of iotaˆ(k) is a lower bound with a median of 0.85 of the true value and no coverage statement, so the regret bound stays a full-table diagnostic and a pilot-side regret certificate is open; milestone 40, `kth_inflate_results.txt`: no fixed multiplier fixes it, since 3 covers 44 of 55 cells at 80% and a multiplier chosen on nine designs covers 44 of the 55 held-out cells, with none finite in 2 cells where the pilot value is 0; milestone 41, `quantile_g_results.txt`: replacing the maximum by the 0.9-quantile of the per-candidate excess gives a bound under the spread of y in 38 of 40 cells that realised regret exceeds in 5.00% of draws, but all in 2 cells where the optimum itself has a large excess, so it is evidence, not a certificate; milestone 43, `quantile_counterexample_results.txt`: no quantile q < 1 of that excess can be a theorem, since a constructed pool has the optimum ranked last by the rung and every other excess 0, so the quantile is 0 and the regret is log c for any n, and the empirical coverage stays a property of this table).

E. 5 Limitations and threats

Ten small designs, one FIFO, one corner, one tool version for the headline tables (milestone 48 re-measured the ten-design table under OpenROAD 26Q3 / Yosys 0.70: the qualitative conclusions reproduce, R4 scores correlate across versions at only tau 0.44 median and the tight clock target moved in 16 of 20 cells, `experiments/rung_bias_26q3/`), pools of about 50 so every strategy converges to 0 regret, global-route estimates, one proof solver and timeout. Pilots are one third of the pool at m = 16. Coverage is over pilots and tie-breaks within a fixed group, not over designs. Hyperband was run in its simplest form (one bracket per start rung, equal budget split; `experiments/hyperband/`): it does not beat random search at equal wall-clock on these pools but halves the worst-design regret of SH from R2 at the relaxed clock (4.6% vs 11.8% at 45 s). No BOHB baseline yet; the smallest model-based baseline we ran, a greedy ridge regression of routed cost on R2 features refitted as points are routed (`experiments/mf_surrogate/`, milestone 20), is worse than random R4 and than never cutting on R2 at budgets of 15, 25 and 40 routed-point equivalents (mean regret 4.8, 3.1, 1.7% against 3.1, 1.4, 0.4% for the latter), and stays at 19% on barrel64; adding an exploration term (a lower confidence bound or epsilon-greedy, milestone 21,

`experiments/mf_explore/`, run once before it was logged) improves every variant over greedy, about halving barrel64's regret at 40 equivalents, but none beats never cutting on R2 or random R4 at any budget (mean regret 4.5, 2.8, 1.3% for the best lower-confidence-bound variant against 3.1, 1.4, 0.4% for the former). A GP Bayesian-optimisation baseline (exact GP with an RBF kernel on the R2 features, expected improvement or lower confidence bound, fixed untuned hyperparameters; milestone 22, `experiments/mf_gp/`, 100 seeds) closes barrel64 at 40 equivalents (0.13%) but still loses to never cutting on R2 at every budget (best mean regret 3.8, 1.9, 0.74% against 3.0, 1.5, 0.45%), losing on 23 of 40 cells at 25 and 40 equivalents. BOHB's model, a TPE density ratio over the same features inside the same single-pool search (milestone 23, `experiments/mf_tpe/`, 100 seeds; the bracket schedule is not reproduced), is weaker than the GP: best mean regret 4.3, 2.3, 1.1% against 3.0, 1.5, 0.45% for never cutting on R2, and 7% on barrel64 at 40 equivalents. NSGA-II likewise. Several effects rest on a handful of designs; there is no multiple-comparison correction.

Tool-version sensitivity of the fanout result (milestones 73 to 76, `experiments/fanout_sweep/`). On the same 60 pass sequences, R2 pick regret on the width-parameterised one-hot mux is 16 / 19 / 27 / 109% (W = 24, 48, 96, 192) under Yosys 0.33 and 0 / 0 / 0 / 16% under Yosys 0.70, both with OpenROAD 26Q3: with the sample fixed, the tool version alone moves pick regret by up to 93 points. What survives both versions is the dose-response of the rank correlation of period (falling with W) and of the routed-to-mapped period ratio (rising). A trend of pick regret with fanout is therefore not claimed as a property of the flow. Two intermediate releases (0.47, 0.51) split the flow's `abc -g` gate list at the commas when it is passed with `-p`; with the commands in a script file (`-s`, which reproduces the 0.70 table exactly) they run and give identical results to each other: pick regret 1.4 / 0 / 0 / 0.9% and tau of period 0.33 / 0.52 / 0.03 / -0.04, so at W = 192 pick regret is 109, 16 and 1% for Yosys 0.33, 0.70 and 0.47/0.51 (milestone 77). The mapped period carries no rank information at W = 192 under any version (tau between -0.04 and 0.10).

F. 6 What the paper still needs

Findings that used to be listed here now sit in Sections 3 and 4. What is open: 1. Full detail-routed cells for mult24 and mult32 (about 12 core-hours per 50-point cell; only 12 and 4 spread points so far) and an OpenROAD run of the larger designs at the ss corner (liberty-only so far). 2. A causal fanout result that does not depend on the Yosys version (Section 5). 3. BO/BOHB baselines on larger pools and with structurally unfaithful R2 beyond the multiplier family. 4. A pilot-side regret certificate (a proof or counterexample for the top-vs-all coverage; Section 4 says what is excluded). 5. Typesetting to the MLCAD template.

Until (1) to (3) land, the honest headline is the negative one: on small designs, cheap rungs did not pay, and a pilot gate that detects when they will not has not been shown to pay for itself on the 30 designs it was replayed on (20 in milestone 16, five each in milestones 27 and 28).